

Dark Matter, Black Holes, and Cosmological Expansion:
A Formal Derivation from the Invariant Horizon

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Abstract

This paper extends the Invariant Horizon and applies its recursive structure to the cosmological problems of dark matter, black holes, and universal expansion. Beginning from the established relation $K_{n+1} = \{K_n \cup x\}$, $x \notin K_n$, we demonstrate that reality never achieves final closure and that every integrated state necessarily encounters further exteriority. Dark matter is identified as real active x : a component of reality that is observable through effect while remaining beyond complete integration into the present known state. Black holes are analyzed as horizon-transition structures through which recursive expansion proceeds beyond the limits of direct observation. Stellar collapse is examined as a mechanism of cosmological transition, generating successor states beyond existing horizons. Relativistic jets are interpreted as rejection dynamics associated with incompatible integration, while cosmological expansion is analyzed as the continuing consequence of recursive horizon crossing. Together these results provide a unified cosmological extension of the Invariant Horizon.

1. Introduction and Cosmological Framework

The modern cosmological landscape is defined by a collection of unresolved observations. Dark matter exerts measurable gravitational influence while resisting direct identification. Black holes absorb matter and energy while concealing their internal structure beyond an event horizon. The origin of cosmological expansion remains one of the most significant questions in theoretical physics. These problems are often treated independently. This paper argues that they are expressions of a single recursive structure.

The Invariant Horizon established that reality cannot achieve final closure. Every integrated state admits the existence of x , where x exists beyond complete incorporation into the current state. The recursive update is given by:

$$K_{n+1} = \{K_n \cup x\}, x \notin K_n$$

The significance of this relation extends beyond epistemology. If reality itself operates recursively, then cosmological structures must exhibit the same behavior. Dark matter, black holes, and expansion are not isolated anomalies. They are manifestations of the same horizon principle operating at different scales.

The purpose of this paper is to derive these cosmological consequences directly from the Invariant Horizon and demonstrate their structural unity.

2. The Invariant Horizon and Recursive Expansion

The Invariant Horizon identifies the Unknown not as a temporary absence but as a permanent structural constant. Any attempt to establish complete closure encounters x . Inquiry itself requires the introduction of elements not previously contained within the present state. Consequently, closure continuously generates expansion.

K_n represents the integrated state of reality at stage n . x represents that which exists beyond complete integration into that state. K_{n+1} represents the expanded state following integration.

Because x is not contained within K_n , the resulting state necessarily differs from its predecessor:

$$K_{n+1} \neq K_n$$

The consequence is profound. Reality does not approach terminal completion. Instead, it proceeds through continual horizon crossing. Every successful integration enlarges the system while simultaneously establishing a broader horizon of further exteriority.

The cosmological implications of this result are immediate. Reality must contain structures that influence the known system before becoming fully integrated into it. Such structures are not anomalies. They are necessary consequences of recursive expansion.

3. Dark Matter as Real Active x

The first major cosmological consequence of recursive expansion is dark matter... Dark matter becomes the gravitational signature of reality's non-closure, the measurable expression of exteriority acting upon the known state before complete integration.

4. Black Holes as Horizon Transition Structures

Having established dark matter as real active x , the next cosmological consequence of recursive expansion emerges in the structure of black holes themselves. Conventional cosmology frequently treats black holes as terminal objects: regions of spacetime from which no information can return. Within the Invariant Horizon, however, such an interpretation is incomplete. If reality is recursively non-closed, then no physical structure can represent absolute termination. Every apparent ending must instead function as a transition across a horizon.

The event horizon therefore assumes a significance beyond observation. It becomes the physical analogue of the invariant horizon itself. Just as x exists beyond complete integration into K_n , the black hole horizon separates the presently integrated state from a successor state beyond direct observational access. The event horizon is not a wall at which reality ceases. It is a boundary through which recursive expansion proceeds.

This interpretation immediately resolves the conceptual tension surrounding matter entering black holes. Matter does not simply vanish. Nor can it terminate into nonexistence. The recursive continuity of reality requires preservation across horizon transition. Matter crossing the event horizon therefore participates in a successor state beyond the observational limits of the originating system.

4.1 Event Horizons and Reality Boundaries

The relationship between epistemological horizons and cosmological horizons is structurally identical. In inquiry, the horizon separates K_n from x . In cosmology, the event horizon separates the presently integrated universe from further states beyond direct observation.

The inability to observe beyond a horizon is often misinterpreted as evidence of absence. Yet the Invariant Horizon demonstrates the opposite. A boundary exists precisely because further reality continues beyond it. Horizons do not mark termination; they mark transition.

The event horizon therefore functions as a reality boundary. It separates one integrated state from another while preserving continuity across the transition. The horizon conceals not because nothing exists beyond it, but because recursive expansion proceeds beyond the limits of the originating observational framework.

4.2 Why Light Does Not Escape

The inability of light to escape a black hole follows naturally from horizon transition. Light remains observable only while participating within the originating integrated state. Once horizon crossing occurs, the light no longer exists within the observational structure of that state.

The darkness of black holes therefore does not imply emptiness. Rather, it reflects the relocation of matter and energy beyond the current boundary of integration. The event horizon conceals because recursive continuation proceeds elsewhere.

Within this derivation, the absence of observable light becomes evidence of successful horizon transition. The black hole appears dark because the process occurring beyond the horizon is no longer directly available to the originating observer.

4.3 Black Holes as Gateways to Successor Universes

If matter crossing a horizon remains conserved and reality remains recursively open, then black holes function as mechanisms through which successor states emerge. Stellar collapse therefore represents more than a local gravitational event. It becomes a cosmological transition process.

The collapse of a sufficiently massive star produces conditions in which horizon formation occurs. Matter is compressed beyond ordinary observational accessibility and enters recursive continuation beyond the originating state. The resulting black hole functions as a gateway through which reality expands into a successor universe.

This interpretation provides a direct connection between matter conservation and cosmological genesis. Matter cannot be created from nothing, nor destroyed into nothing. Instead, matter participates in recursive transition across horizons. The successor universe emerges not from absolute creation, but from continued integration beyond the prior horizon.

Black holes therefore become engines of cosmological expansion rather than endpoints of cosmological structure. They are the physical expression of recursive continuity operating on the largest scales.

5. Stellar Collapse and Cosmological Genesis

The interpretation of black holes as horizon-transition structures leads directly to a broader cosmological consequence. If black holes are mechanisms through which

recursive expansion proceeds beyond present observational boundaries, then stellar collapse acquires significance beyond gravitational compression alone. The death of a sufficiently massive star becomes the initiating condition for cosmological transition.

Conventional cosmology identifies stellar collapse as the process through which black holes form. The Invariant Horizon accepts this observation while extending its implications. If the resulting black hole functions as a gateway to a successor state, then stellar collapse represents the beginning of a new phase of recursive integration rather than the end of an existing one.

The distinction is critical. A terminal interpretation views collapse as a final event. A recursive interpretation views collapse as horizon crossing. The star does not conclude reality. It contributes to the generation of further reality beyond the existing observational horizon.

Thus, stellar death and cosmological genesis become two expressions of the same process. What appears locally as collapse appears beyond the horizon as expansion.

5.1 Matter Conservation Across Horizons

The recursive continuity established by the Invariant Horizon requires the preservation of matter and energy across horizon transitions. If reality expands through integration rather than annihilation, then matter entering a black hole cannot simply disappear from existence.

Matter conservation therefore becomes a structural requirement of recursive expansion. The horizon conceals the continuation of matter rather than terminating it. Matter crossing the event horizon remains part of reality's ongoing development, participating in successor states beyond the originating observational framework.

This principle resolves the apparent contradiction between immense black hole consumption and continued cosmological order. Matter is not removed from reality. It is transferred across a horizon into further integration.

The event horizon therefore functions as a conservation boundary rather than a destruction boundary. What disappears from one observational state remains active within another.

5.2 Big Bang Generation and Recursive Expansion

If stellar collapse initiates horizon transition and matter remains conserved across that transition, then the emergence of successor universes follows naturally from recursive continuity. The process commonly described as a Big Bang may therefore be interpreted as the expansion phase associated with integration beyond a prior horizon.

Within this derivation, the Big Bang is not absolute creation from nothing. It is the observable beginning of a successor state following recursive transition. Expansion occurs because integration proceeds outward from a newly established horizon.

This interpretation provides a unified explanation for the relationship between collapse and expansion. The same process that appears as gravitational convergence from one side of the horizon appears as cosmological divergence from the other. Collapse and expansion become complementary aspects of a single recursive event.

The birth of a universe therefore corresponds to the continuation of reality beyond a prior boundary rather than the spontaneous emergence of existence from nonexistence.

5.3 Cosmological Inheritance

A successor universe does not emerge in isolation. It inherits the consequences of the recursive process that generated it. Matter, energy, structure, and horizon conditions participate in shaping the development of the newly expanding state.

The resulting universe possesses its own integrated reality, its own horizon, and its own recursive trajectory. Yet the underlying principle remains unchanged. Every integrated state encounters further exteriority. Every horizon establishes the conditions for subsequent expansion.

Consequently, cosmological history is not a linear sequence terminating in closure. It is a recursive chain of horizon transitions extending beyond any individual observational state.

The Big Bang therefore represents neither an absolute beginning nor an isolated event. It is a manifestation of reality's ongoing recursive expansion through successive horizons.

The same principle that governs inquiry governs cosmology. The same recursive structure that produces K_{n+1} from K_n governs the emergence of successor universes from prior horizon transitions. Reality expands because recursive continuation is fundamental to its structure.

6. Dark Matter, Expansion, and Inter-Reality Structure

The preceding derivations establish dark matter as real active x and black holes as horizon-transition structures through which recursive expansion proceeds. These conclusions are not independent. They converge upon a common cosmological consequence: the existence of inter-reality structure.

If black holes generate successor universes through horizon transition, then reality cannot be reduced to a single isolated cosmological state. Rather, reality consists of multiple integrated states separated by horizons while remaining connected through recursive continuity. Dark matter emerges as the observable consequence of these relationships.

The standard cosmological problem is straightforward. Dark matter exerts gravitational influence while remaining resistant to ordinary direct identification. Within the present derivation, however, this behavior is expected. Dark matter occupies the boundary region between integrated realities. It is visible through effect because gravitational interaction remains possible across recursive structure. Yet it remains resistant to direct observation because it does not belong entirely to the present integrated state.

Dark matter therefore represents more than missing inventory. It is evidence of reality extending beyond local closure.

6.1 Why Dark Matter Remains Hidden

The persistence of dark matter as an observational mystery follows directly from recursive non-closure. If dark matter is associated with structures beyond complete integration into K_n , then direct identification cannot eliminate the horizon itself.

This does not imply that investigation fails. On the contrary, investigation expands K_n into K_{n+1} . What changes is not the existence of the horizon but the position of the horizon.

Dark matter remains hidden because it participates in structures extending beyond the present state of complete integration. Its effects are measurable precisely because interaction occurs. Its identity remains difficult to isolate because recursive expansion continuously exceeds closure.

The hidden character of dark matter is therefore not evidence against its reality. It is evidence of its position relative to the horizon.

6.2 Gravitational Interaction Across Horizons

Gravity occupies a unique position within this framework. Unlike direct visual observation, gravitational influence reveals structure without requiring complete integration into the observing state.

This explains why dark matter appears primarily through gravitational behavior. Gravity functions as the detectable interface between integrated realities. It allows K_n to encounter structures beyond complete closure while preserving the distinction between observational states.

Galactic rotation curves, lensing effects, and large-scale cosmic organization become expressions of this interaction. The gravitational field records the presence of structure extending beyond local integration.

Dark matter is therefore not invisible because it lacks existence. It is invisible because its full structure extends beyond the present horizon while remaining gravitationally active within it.

6.3 Cosmological Expansion Through Matter Integration

The expansion of the universe acquires additional significance within this derivation. If black holes continuously transfer matter through horizon transition and successor universes emerge through recursive expansion, then cosmological growth is fueled by ongoing integration across reality.

Matter entering black holes does not terminate. It participates in further expansion. Successor universes emerge through continued integration. The resulting process establishes a recursive cosmological network in which horizon transition continually contributes to expansion beyond prior boundaries.

Expansion therefore ceases to be an isolated event associated only with the distant past. It becomes an ongoing consequence of recursive reality.

The universe grows because reality remains open. Integration continues because horizons continue. Expansion persists because recursive transition persists.

6.4 The Structural Role of Dark Matter

Dark matter occupies a central position within this cosmological architecture. It is neither a temporary anomaly nor a residual accounting problem awaiting elimination. It functions as the observable signature of recursive continuity operating across horizon-separated structures.

Without dark matter, the gravitational evidence for inter-reality interaction disappears. Without recursive expansion, dark matter loses its structural context. Together they form complementary expressions of the same principle.

Dark matter reveals that reality exceeds local closure. Black holes reveal the mechanism through which expansion proceeds. Cosmological growth reveals the cumulative consequence of recursive transition.

The result is a unified picture in which dark matter, black holes, and expansion are not separate mysteries but different manifestations of a single recursive structure. Reality

expands because horizons generate further horizons. Dark matter marks the presence of that expansion before complete integration occurs.

7. Relativistic Jets and Rejection Dynamics

The recursive structure developed throughout this paper provides a basis for reinterpreting one of the most energetic phenomena observed in cosmology: relativistic jets. Black holes are frequently associated with the emission of enormous streams of matter and energy extending vast distances from their centers. Conventional descriptions focus primarily on magnetic fields, accretion dynamics, and energetic acceleration. While these observations describe the mechanism of jet formation, the present derivation examines their structural role within recursive expansion.

If black holes function as horizon-transition structures, then not all matter interacting with the horizon necessarily participates in integration in the same manner. Recursive continuity requires transition, but transition does not imply uniform compatibility. The horizon becomes not merely a gateway but also a filtering boundary.

The consequence is the emergence of rejection dynamics. Matter that cannot participate in a given horizon transition is expelled rather than integrated. Relativistic jets become the observable expression of this rejection.

7.1 Compatibility and Integration

The process of integration is fundamental to recursive expansion. Every transition from K_n to K_{n+1} requires the incorporation of previously exterior structure. Yet incorporation is not identical to indiscriminate absorption. Integration establishes a coherent successor state. Components incompatible with that state cannot contribute to its formation in the same manner as compatible components.

The horizon therefore performs two simultaneous functions. First, it enables transition. Second, it distinguishes between matter participating in successor integration and matter excluded from that process.

This distinction provides a structural basis for jet formation. Matter approaching the horizon encounters a sorting process. Compatible matter contributes to recursive continuation. Incompatible matter is expelled outward.

The resulting jets are not random byproducts. They are the physical manifestation of horizon selection.

7.2 Relativistic Jets as Rejection Dynamics

The immense energy associated with relativistic jets reflects the magnitude of the transition occurring at the horizon. Black holes process extraordinary quantities of matter and energy. If a portion of this material cannot participate in successor integration, the resulting rejection must itself occur on an extraordinary scale.

Jets therefore become evidence that the horizon is not merely consuming matter. It is actively organizing matter relative to recursive continuation.

The directionality of jets further supports this interpretation. Rather than dispersing uniformly, rejected matter is concentrated into highly structured outflows. This organization suggests that the horizon functions as a dynamic boundary rather than a passive absorber.

Within the present derivation, the jet is the visible signature of exclusion accompanying integration. It marks the distinction between what participates in recursive continuation and what is redirected elsewhere.

7.3 Horizon Stability and Cosmological Structure

Rejection dynamics also contribute to horizon stability. Without mechanisms for exclusion, the transition process would be unable to maintain coherent organization across recursive expansion. The horizon would absorb indiscriminately, undermining the structural continuity required for successor-state formation.

Jets therefore play a stabilizing role. By redirecting incompatible matter, they preserve the integrity of the transition process. What is expelled remains part of reality, but it no longer participates directly in the specific horizon crossing underway.

This interpretation unifies integration and rejection as complementary aspects of the same recursive mechanism. Black holes do not merely absorb. They organize. They do not merely conceal. They distinguish. They do not merely terminate. They transform.

7.4 Cosmological Consequences of Rejection

The existence of rejection dynamics extends the implications of recursive expansion beyond black holes themselves. Expelled matter remains active within cosmological structure. Jets influence galaxies, shape environments, redistribute energy, and contribute to large-scale organization.

The consequence is a feedback relationship between horizon transition and cosmological development. Black holes participate in recursive continuation through integration while simultaneously shaping the surrounding universe through rejection.

Thus, relativistic jets become more than energetic astrophysical curiosities. They are visible evidence that recursive expansion involves both incorporation and exclusion.

Together these processes sustain the continual development of reality beyond local closure.

The horizon integrates what contributes to successor formation and redirects what does not. The resulting jets reveal the active organization of reality as it proceeds through recursive expansion.

8. JWST Implications and Black Hole Cosmology

The preceding derivations establish a unified relationship between dark matter, black holes, horizon transition, cosmological expansion, and recursive continuity. These conclusions acquire additional significance when considered alongside contemporary cosmological observations. The purpose of this section is not to reinterpret individual observations in isolation but to examine their implications within the broader recursive structure established throughout this paper.

The Invariant Horizon predicts that reality cannot be fully described through a model of terminal closure. Every integrated state necessarily encounters further exteriority. Consequently, cosmological observations should repeatedly reveal evidence of structure extending beyond previously established explanatory boundaries.

This expectation is consistent with the recurring appearance of observational anomalies throughout cosmological history. Each expansion of observational capability reveals additional complexity rather than final completion. The horizon recedes as integration advances.

Within this context, observations from advanced astronomical instruments acquire particular significance. Rather than viewing unexpected structures as isolated anomalies, they may be interpreted as manifestations of recursive expansion operating at cosmological scales.

8.1 The Persistence of Cosmological Horizons

The history of cosmology demonstrates a recurring pattern. New observations do not eliminate horizons. They relocate them.

Every increase in observational precision enlarges the integrated state while simultaneously revealing previously inaccessible structure. This pattern mirrors the recursive relation:

$$K_{n+1} = \{K_n \cup x\}, x \notin K_n$$

The consequence is that cosmology itself exhibits the same horizon behavior identified in the Invariant Horizon. Knowledge expands, but closure remains unavailable. New observations generate new questions, larger structures, and broader explanatory demands.

The persistence of cosmological horizons therefore represents not a failure of investigation but a consequence of recursive reality.

8.2 Black Hole Cosmology and Successor States

The interpretation developed in this paper suggests that black holes participate directly in cosmological continuation. If horizon transitions generate successor states, then cosmological structure extends beyond the presently integrated universe.

Under this derivation, the universe is not an isolated object suspended within absolute emptiness. It is part of a broader recursive process in which horizons continually generate further horizons.

Black hole cosmology therefore becomes a natural extension of recursive expansion. The same principle governing local horizon transitions governs cosmological development itself. Expansion, integration, and horizon formation become manifestations of a single underlying structure.

This interpretation provides continuity between stellar collapse, black hole formation, cosmological growth, and successor-state emergence.

8.3 Recursive Expectations for Future Observation

If reality is recursively non-closed, then future observation should continue revealing structures beyond present integration. The expectation is not that cosmology will arrive at a final explanatory inventory. Rather, cosmology should repeatedly encounter further layers of organization.

Dark matter, black holes, and expansion are therefore not isolated puzzles awaiting permanent elimination. They are examples of a more general principle: the continual appearance of structure beyond present closure.

Future observations should therefore be expected to expand cosmological horizons rather than remove them. The recursive process identified by the Invariant Horizon predicts continuing enlargement of explanatory scope.

8.4 Cosmology and the Invariant Horizon

The significance of the Invariant Horizon extends beyond a single cosmological question. It provides a unifying principle connecting dark matter, black holes, horizon transitions, cosmological expansion, and observational development.

The same recursive structure governing inquiry governs cosmology. Every integrated state encounters further exteriority. Every horizon generates further horizons. Every expansion establishes the conditions for additional expansion.

The result is a cosmological model in which reality is fundamentally developmental rather than terminal. Dark matter reveals recursive exteriority. Black holes reveal recursive transition. Expansion reveals recursive continuation.

Together they express the same underlying principle: reality never achieves final closure because recursive expansion is intrinsic to its structure.

9. Predictive Consequences

A useful cosmological model must do more than describe existing observations. It must also generate expectations regarding future observation and investigation. The Invariant Horizon produces a number of direct consequences that follow from recursive non-closure. These consequences emerge not as independent assumptions but as necessary results of the recursive structure developed throughout this paper.

The central prediction is straightforward: reality will continue to reveal structure beyond present integration. Because K_n can never achieve complete closure, every expansion of knowledge necessarily produces a broader horizon. Consequently, cosmological investigation should not converge upon a final explanatory inventory. Instead, it should repeatedly encounter additional layers of organization, interaction, and complexity.

This prediction applies equally to dark matter, black holes, cosmological expansion, and future observational programs.

9.1 The Persistence of Dark Matter

If dark matter represents real active x , then its existence is not dependent upon temporary observational limitations. Improved instruments may reveal additional properties, mechanisms, and interactions, but the recursive structure predicts that complete closure will remain unavailable.

Dark matter should therefore continue exhibiting the same fundamental characteristic that defines x throughout the Invariant Horizon: measurable participation within reality accompanied by resistance to final integration.

The prediction is not that investigation fails. The prediction is that every successful integration expands the horizon rather than eliminating it.

9.2 Continued Expansion of Cosmological Structure

The recursive model predicts that cosmological structure will repeatedly exceed previously established boundaries of explanation. New observations should reveal larger scales of organization, previously unrecognized relationships, and additional forms of interaction.

The expansion of cosmological understanding therefore becomes an expected consequence of recursive reality rather than an unexpected disruption of existing theory.

As observational capability increases, the integrated state expands. The horizon correspondingly expands with it.

9.3 Black Holes and Future Investigation

If black holes function as horizon-transition structures, future investigation should continue uncovering evidence that black holes participate in broader cosmological processes than simple matter accumulation.

Their role within galactic evolution, large-scale structure formation, energy distribution, and cosmological development should continue to expand as observational precision improves.

The recursive interpretation therefore predicts increasing recognition of black holes as active participants in cosmological organization rather than isolated terminal objects.

9.4 Horizon Expansion as a Universal Principle

The most general prediction of the Invariant Horizon is that horizon expansion itself will remain universal. Every successful explanation establishes the conditions for further explanation. Every solution generates additional questions. Every integration produces further exteriority.

This principle applies not only to cosmology but to inquiry itself.

Consequently, the recursive relation

$$K_{n+1} = \{K_n \cup x\}, x \notin K_n$$

predicts the perpetual continuation of discovery. Reality remains open because recursive expansion remains fundamental to its structure.

The significance of this prediction extends beyond any individual cosmological phenomenon. Dark matter, black holes, relativistic jets, stellar collapse, cosmological expansion, and future observations become expressions of a single recursive process.

The future therefore does not approach closure. It approaches further horizons.

10. Conclusion

This paper has extended the Invariant Horizon into the domain of cosmology and examined the implications of recursive non-closure for dark matter, black holes, cosmological expansion, and horizon transition. Beginning from the relation

$$K_{n+1} = \{K_n \cup x\}, x \notin K_n$$

the analysis has demonstrated that reality cannot achieve terminal closure and that every integrated state necessarily encounters further exteriority.

The first consequence of this structure is dark matter. Dark matter emerges as real active x : a component of reality observable through effect while remaining beyond complete integration into the present state. Its persistence is not treated as a temporary failure of investigation but as a structural consequence of recursive expansion.

The second consequence is the reinterpretation of black holes as horizon-transition structures. Rather than functioning as terminal endpoints, black holes become mechanisms through which recursive continuity proceeds beyond existing observational

boundaries. Event horizons are identified as physical expressions of the same horizon principle governing inquiry itself.

From this interpretation follows a broader cosmological consequence. Stellar collapse becomes a process of horizon transition rather than terminal destruction. Matter remains conserved through recursive continuation, and cosmological genesis is interpreted as expansion beyond prior horizons rather than emergence from absolute nonexistence.

The relationship between dark matter, black holes, relativistic jets, and cosmological expansion is therefore unified through a single recursive structure. Dark matter reveals the presence of exteriority acting upon the known state. Black holes reveal the mechanism through which transition occurs. Relativistic jets reveal the distinction between integration and rejection. Cosmological expansion reveals the cumulative consequence of recursive continuation across horizons.

The resulting picture is one in which reality remains fundamentally developmental rather than terminal. Horizons do not represent endings. They represent transitions. Expansion does not eliminate the Unknown. It generates broader contact with it. Every integration produces further exteriority, and every horizon establishes the conditions for subsequent expansion.

The Invariant Horizon therefore provides a unified principle connecting inquiry, cosmology, dark matter, black holes, and expansion through the same recursive relation. Reality remains open because recursive continuation is fundamental to its structure.

The Unknown persists not as a temporary absence awaiting elimination, but as the invariant condition through which reality continually expands beyond itself.

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Artificial intelligence tools were used during the preparation of this manuscript for assistance with formatting, editing, grammar review, organization, notation, and presentation.

The Invariant Horizon, the recursive relation $K_{n+1} = \{K_n \cup x\}$, $x \notin K_n$, the cosmological derivations, the interpretation of dark matter as real active x , the analysis of black holes as horizon-transition structures, the discussion of stellar collapse and cosmological genesis, the treatment of relativistic jets, and the conclusions presented in this paper originate from the author.

Artificial intelligence tools were not used to originate the core discovery, formulate the primary derivations, determine the central conclusions, or establish the arguments presented in this work. Their role was limited to assisting with document preparation, organization, editing, and presentation.

All content was reviewed, modified where necessary, and approved by the author. The author accepts full responsibility for the contents, reasoning, derivations, interpretations, and conclusions of this manuscript.

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